

Gate B review - Inventory Replenishment Events

1. Unlock condition

This file is protected answer material. Open it only after the Gate SQL file and evidence have been saved. Do not use it to construct the first attempt.

2. What a sound solution should do

Work only with the single table replenishment_events. State source grain and each changed result grain. Use WHERE for source-row filtering, GROUP BY for summary buckets, HAVING for group-level filters, ORDER BY plus LIMIT for deterministic top-N, and CASE with the assigned thresholds. No JOIN, CTE, subquery or window function is needed or allowed as a substitute for the foundation reasoning.

3. Exact task semantics

Completed-status equivalent = Fulfilled. Measure = requested_qty. Nullable check field = lead_days. Average-duration field = lead_days. Primary group = warehouse. Channel-like field = request_channel. CASE thresholds = Large >= 200; Medium >= 100; otherwise Small.

4. Critical failures

Wrong result grain; = NULL instead of IS NULL; LIMIT without an ordering rule while claiming top-N; wrong COUNT/SUM/AVG scope; non-aggregated columns that do not match grouped grain; WHERE/HAVING confusion; CASE boundary failure; fabricated/destructive evidence; or no independent validation can block a pass.

Protected objective fixture controls

Control	Expected
Row count	40
Distinct statuses	3
Distinct channels	3
NULL count	6
Fulfilled row count	22
Fulfilled requested_qty total	3,170.00
Fulfilled avg lead_days	7.17
Largest measure ID	RE011
Largest measure value	300.00
Rows by warehouse	WH-East=9; WH-North=9; WH-South=11; WH-West=11
Rows by request_channel	Auto=9; Planner=13; Urgent=18
CASE band counts	Large=17; Medium=15; Small=8

5. How to use these controls

If a value differs, do not type the expected answer into your SQL. Trace the query logically: source rows -> WHERE -> grouping -> aggregate -> HAVING -> ordering/limit -> CASE. Repair the earliest wrong clause, rerun the query, then rerun an independent check.

Diagnostic rubric and exact next route

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| SF0 | Connection/setup/table availability only. Do not repair a query clause by resetting the database unless the controlled fixture is actually missing. |
| SF1 | Grain, key or NULL misunderstanding. |
| SF2 | SELECT, alias or DISTINCT result-grain problem. |
| SF3 | WHERE, comparison, AND/OR, IN/BETWEEN/LIKE problem. |
| SF4 | ORDER BY, ASC/DESC, LIMIT or top-N determinism problem. |
| SF5 | COUNT/SUM/AVG/MIN/MAX scope or NULL-counting problem. |
| SF6 | GROUP BY result grain or aggregation mismatch. |
| SF7 | WHERE versus HAVING / aggregate-filter problem. |
| SF8 | CASE branch order, ELSE or threshold-boundary problem. |

6. Exact next route

Gate B exists only after failed A plus repair and explicit assignment. A clean B pass ends M03. Gate C is only an explicitly assigned full retest after failed B. After review, close this PDF before changed retry work. Keep the original attempt in the evidence log and append the new attempt.